Study Project

Simpson's Approach for Geometric Nonlinear Sensitivity

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Abstract

As the world is moving more towards sustainable development, Structural optimization has become more essential field. While performing optimization with gradient-based methods, the computation of accurate sensitivity is critical as it gives information on the importance of a design variable in a system. In most existing research on structural optimization for geometric nonlinear problems, its sensitivity is calculated with linearized end compliance. This study project proposes an alternative objective formulation based on Simpson's integration rule, which is capable of capturing geometric nonlinearity. First, the objective function with Simpson's rule is formulated for geometric nonlinear strain energy. Its sensitivity formulation is carried out, and the respective analytical sensitivity is derived. Results show that this method can significantly reduce the computational cost by 10 times when compared to the trapezoidal approach and can increase accuracy by 50% when compared to Linear end compliance sensitivity.

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1. Literature Review

The first research on optimization with stability started in 1970's for beam elements [1]. The first work of non-linear shape optimization for shell elements appeared in 1995 [2]. They investigated on sensitivity analysis of shell structures. The work by Y.X. Gu also addressed the same, and also had results for shell optimization for buckling, but it lacked good documentation of results for shell elements [3]. The implementation code was also missing. The work by Prof. Bletzinger in 1994 provided a framework for shape optimization to increase the buckling load [4]. Although the method is a general one, the results involved only 2D structures and left the scope of 3D shape optimization. One of the significant works in shape optimization with proper FEM implementation for shell elements is by Mathias Firl [5]. Following his work by Wenjia Wang was also notable one. In topology optimization, the work by T. Buhl in 2000 was the first formulation for Geometric nonlinear topology optimization [6].

In these literatures, the objective function for nonlinear compliance is approximated and simplified. Instead of tracing the actual nonlinear load displacement graph and finding the area under it, a linear load displacement curve for the specific load value (for a specific tangent stiffness matrix, K) is taken, and compliance for it is found, termed as end compliance. The above-mentioned end compliance is also discussed by Bendose in his book on optimization, page 88 [7].

This simplifies a good amount of computational cost as nonlinear displacements are computed only once for one iteration (for all elements). But this approach introduces a problem with the optimized shape. It is clearly discussed at [8], [9]. By linearizing the nonlinear compliance, we are neglecting information about the nonlinear behaviour before the load point. This results in optimizing the structure only for a specific load case, which might end in a non-optimized structure if the load value is not chosen properly [8]. For example, if the applied load is post-critical point for the structure, and if we optimize the structure for this load, now instead of increasing the critical buckling load, the critical buckling load will be decreased, but the post-buckling strength will be increased [6][9]. Here, although the post buckling strength is increased, it is not our point of interest. Hence, consideration of actual strain energy gains importance. One of the solutions provided by T. Buhl was using multiple objective functions with different load end compliance or using the trapezoidal rule of approximation for compliance computation.

Some of the improvement strategies performed and verified in topology optimization are given. The work by J. Stegmann, where multiple load inputs were given on a snap-through structure, a new objective function was formed with non-linear end compliance for each of these loads and optimized. The rule for the formation of the weightage function is shown in his paper. It can be seen a good improvement in the overall load displacement curve by choosing 10 load values.

In shape optimization, the master's thesis by Mahmoud Sesa has implemented the trapezoidal rule in shape optimization and validated sensitivities against numerical values. Although the trapezoidal approach and the multiload approach produce better results than end compliance, they require much higher computational cost because many numerical simulations have to be run to get an accurate tracing of the nonlinear load displacement curve.

2. Study Project Proposal:

This study project proposes an alternative approach based on Simpson's rule to analytically calculate the geometric nonlinear sensitivity. First, the analytical sensitivity formulation for the Simpson's approach is derived. This approach is applied to the Euler-Bernoulli beam element with one of the cross-sectional properties as a design variable. This analytical sensitivity is validated against numerically obtained sensitivity values with the trapezoidal approach. The computations are performed on self-made software, which can perform second-order calculations with geometric stiffness matrices. This document is organised as follows

1. First, the objective function is formulated using Simpson's approach.
2. Then the analytical sensitivity equation is derived for Simpson's approach.
3. For reference sensitivity values, finite difference sensitivity is carried out with a trapezoidal approach on the actual load displacement curve, and this is compared against the finite difference sensitivities of the Simpson's approach.
4. Then, the analytical sensitivity values of Simpson's are compared against its numerical sensitivity (to prove the formulation is correct) and against the reference value.
5. Inference
6. References

3. Objective Formulation

Area under a quadratic curve by Simpsons = $1/6 * x * (4y_1 + y_2)$

$y_2 = f(x_2)$

$y_1 = f(x_2/2)$

Here, the rule is that y_1 must be found at the midpoint of x. If we take y as load and x as displacement, U, then it is impossible to match $U_1 = U_2/2$ for all U values. Hence, x is taken as load and y is taken as displacement, by which the area under the curve becomes.

Area = $1/6 * F_2 * (4U_1 + U_2)$,

Where U_2 is the second-order displacement for force F_2

U_1 is the second-order displacement for force $F_1 = F_2/2$.

This area is complementary strain energy since we have taken x and y axes oppositely. Hence

$$\textbf{Actual strain energy} = F_2U_2 - 1/6 * F_2 * (4U_1 + U_2) \qquad \textbf{- Equation 1}$$

4. Analytical Adjoint Sensitivity Formulation

Complimentary dJ/dS = $\left[\frac{4}{6} * F_2^T * K_1^{-1} * \frac{dK1}{dS} * U1\right] + \left[\frac{1}{6} * F_2^T * K_2^{-1} * \frac{dK2}{dS} * U2\right]$

Adjoint analytical sensitivity equation,

Complimentary dJ/dS = $\left[\frac{4}{6} * (K_1^{-1}F_2\ )^T * \ \frac{dK1}{dS} * U1\right] + \left[\frac{1}{6} * (K_2^{-1}F_2\ )^T * \frac{dK2}{dS} * U2\right]$

$$\mathbf{dJdS} = \left[\boldsymbol{U_2^T} * \frac{\boldsymbol{dK2}}{\boldsymbol{dS}} * \boldsymbol{U_2}\right] - \left[\left[\frac{\boldsymbol{4}}{\boldsymbol{6}} * (\boldsymbol{K_1^{-1}}F_2\ )^{\boldsymbol{T}} * \ \frac{\boldsymbol{dK1}}{\boldsymbol{dS}} * \boldsymbol{U1}\right] + \left[\frac{\boldsymbol{1}}{\boldsymbol{6}} * (\boldsymbol{K_2^{-1}}F_2\ )^{\boldsymbol{T}} * \frac{\boldsymbol{dK2}}{\boldsymbol{dS}} * \boldsymbol{U2}\right]\right]$$

-Equation 2

5. Finite Difference Sensitivity Comparison

The following two structures (Fig. 1, 2) are studied throughout this study to apply and validate the proposed methodology.

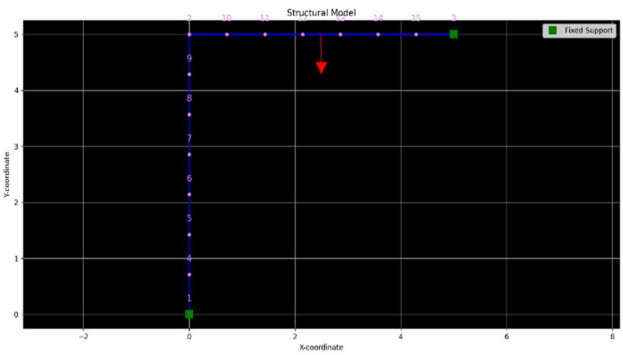

Fig. 1: Model 1

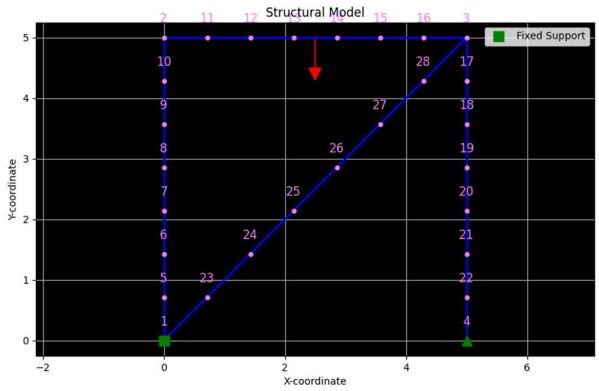

Fig. 2: Model 2

5.1 Model 1 - Description:

Support condition – Both supports are fixed in all 3 Dofs
Discretization – 7 finite elements for each beam
Area – 0.09
Moment of inertia – 0.000675

5.1 Model 2 - Description:

Support condition – Bottom supports – Fixed and Hinged
Discretization – 7 finite elements for each beam
Area – 0.09

Moment of inertia – 0.000675

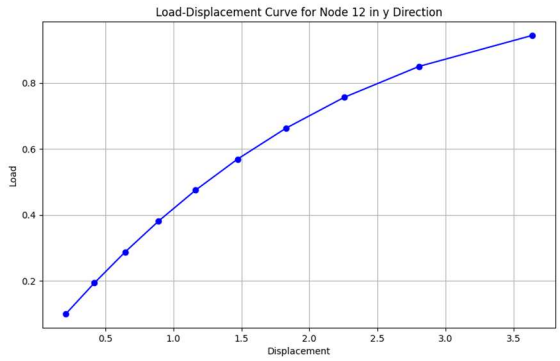

Fig. 3: Load Displacement Curve – Model 1

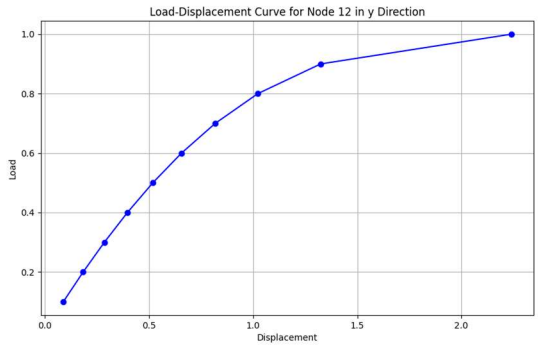

Fig. 4: Load Displacement Curve – Model 1

Figures 3 and 4 give the load displacement curves for Models 1 and 2 in the Y-direction for node 12, respectively.

It shall be observed that both follow a nonlinear load displacement curve; hence, they are considered as appropriate candidates for this study.

5.3 Computation of Finite Difference Sensitivity – with Trapezoidal approximation

1. The load displacement curve is computed with 10 curve points with the second-order Bernoulli beam element
2. The objective value computed by computing the area under this load displacement using the trapezoidal rule
3. Then the design variable is increased by 1e-12, and again the objective is calculated.
4. The change in this objective value with respect to the change in the design variable is the sensitivity

The table below (Tables 1 and 2) shows the sensitivity of each design variable for models 1 and 2

Element	Trapezoidal Numerical Sensitivity
1	-1.104798e-01
2	-7.375213e-05
3	-3.249315e-05
4	-1.104082e-01
5	-1.106030e-01
6	-1.105391e-01
7	-1.102935e-01
8	-2.706809e-04
9	-2.637880e-04
10	-1.108357e-01

Element	Trapezoidal Numerical Sensitivity
1	-5.351950e-05
2	-1.256424e-05
3	-4.382920e-06
4	-3.702141e-05
5	-7.852411e-05
6	-8.787820e-05
7	-5.498156e-05
8	-8.172472e-06
9	-2.788368e-05
10	-1.296183e-04

11	-1.109194e-01
12	-1.103296e-01
13	-1.252091e-04
14	-7.999775e-04

Table 1: Trapezoidal Numerical Sensitivity Model 1

11	-2.322649e-04
12	-1.538284e-04
13	-4.883396e-05
14	6.916816e-07
15	-6.916508e-05
16	-2.364756e-05
17	-2.139638e-04
18	-4.391713e-04
19	-4.623701e-04
20	-2.598006e-04
21	-4.549429e-05
22	-1.355115e-05
23	-2.371264e-06
24	-1.544249e-07
25	-2.387096e-06
26	-1.390080e-05
27	-5.788187e-05
28	-2.188360e-04

Table 2: Trapezoidal Numerical Sensitivity Model 1

5.4 Computation of Finite Difference Sensitivity – with Simpson's formulation

1. Two points on the load displacement curve are evaluated with the second-order Bernoulli beam element. (Note: The Second point is the midpoint of zero displacement and end displacement)
2. The objective values are computed with the proposed Simpson's formula.
3. Then the design variable is increased by 1e-12, and again the objective is calculated.
4. The change in this objective value with respect to the change in the design variable is the sensitivity

Element	Simpsons Numerical Sensitivity
1	-9.693171e-02
2	-6.935804e-05
3	-3.043236e-05
4	-9.686407e-02
5	-9.704776e-02
6	-9.698858e-02
7	-9.675711e-02
8	-1.625775e-05
9	-2.469005e-04
10	-9.726901e-02
11	-9.735355e-02
12	-9.679295e-02
13	-1.183794e-04
14	-7.613202e-04

Element	Simpsons Numerical Sensitivity
1	-4.921336e-05
2	-1.155457e-05
3	-4.011265e-06
4	-3.396929e-05
5	-7.223693e-05
6	-8.116016e-05
7	-5.111302e-05
8	-7.652445e-06
9	-2.578314e-05
10	-1.205698e-04
11	-2.165848e-04
12	-1.419382e-04
13	-4.418183e-05
14	4.619942e-07
15	-6.041443e-05

Table 3: Simpsons Numerical Sensitivity Model 1

16	-2.207718e-05
17	-1.932114e-04
18	-3.938280e-04
19	-4.134690e-04
20	-2.320080e-04
21	-4.060339e-05
22	-1.253810e-05
23	-2.213579e-06
24	-1.472072e-07
25	-2.270506e-06
26	-1.297920e-05
27	-5.314877e-05
28	-1.982082e-04

Table 4: Simpsons Numerical Sensitivity Model 1

The table below (Tables 3 and 4) shows the sensitivity calculated with Simpson's approach (Finite difference) for each design variable for models 1 and 2

5.5 Error in numerical sensitivity

Tables 5 and 6 show the error (in percentage) for Simpson's approach (Finite Difference) and Linearized end compliance (Finite Difference) against the Trapezoidal approach (Finite Difference).

Element	Simpsons Error percentage (%)	Linearized Error percentage (%)
1	12.26	47.35
2	5.96	40.93
3	6.34	42.19
4	12.27	47.36
5	12.26	47.34
6	12.26	47.34
7	12.27	47.35
8	9.88	30.12
9	6.40	40.79
10	12.24	47.31
11	12.23	47.27
12	12.27	47.35
13	5.45	36.69
14	4.83	34.14
Overall	**9.78**	**43.11**

Table 5: Simpsons Numerical Sensitivity Model 1

Element	Simpsons Error percentage (%)	Linearized Error percentage (%)
1	8.05	42.27
2	8.04	42.32
3	8.48	43.51

4	8.24	42.93
5	8.01	42.21
6	7.64	40.77
7	7.04	36.78
8	6.36	25.97
9	7.53	40.53
10	6.98	37.82
11	6.75	36.47
12	7.73	39.78
13	9.53	44.25
14	33.21	439.78
15	12.65	46.16
16	6.64	37.17
17	9.70	44.93
18	10.32	45.80
19	10.58	46.09
20	10.70	46.22
21	10.75	46.27
22	7.48	40.29
23	6.65	38.57
24	4.67	33.08
25	4.88	34.12
26	6.63	38.62
27	8.18	41.96
28	9.43	44.17
Overall	**9.03**	**54.96**

Table 6: Simpsons Numerical Sensitivity Model 1

It shall be observed that the use of Simpson's approach reduces the error by around 50%

6. Analytical vs Numerical Sensitivity Comparison

By using equation 2, analytical sensitivity is calculated in the software. Tables 7 and 8 below show the analytical sensitivity values for models 1 and 2. The analytical sensitivity matches with numerical sensitivity for Simpson's approach. Hence, the proposed analytical sensitivity formulation is valid.

Element	Analytical Simpsons Value
1	-0.09634729877518387
2	-0.00006936031866048564
3	-0.00003040507801783534
4	-0.09641490458583223
5	-0.09623122893239777
6	-0.09629040657490799
7	-0.09652188270311363
8	-0.00001626966278579 18
9	-0.00024691357891310525
10	-0.09600997964136142
11	-0.097355248673321

Element	Analytical Simpsons Value
1	-4.920121e-05
2	-1.156036e-05
3	-4.006517e-06
4	-3.396404e-05
5	-7.223161e-05
6	-8.115158e-05
7	-5.110109e-05
8	-7.674845e-06
9	-2.578176e-05
10	-1.206364e-04
11	-2.166003e-04

12	-0.09648608241470774
13	-0.0001183749122734725
14	-0.0007613194557468739

Table 7: Analytical Simpsons Sensitivity Model 1

12	-1.419485e-04
13	-4.420706e-05
14	4.523340e-07
15	-6.039876e-05
16	-2.210129e-05
17	-1.932239e-04
18	-3.938293e-04
19	-4.134556e-04
20	-2.320061e-04
21	-4.059748e-05
22	-1.253713e-05
23	-2.209056e-06
24	-1.393886e-07
25	-2.262527e-06
26	-1.297078e-05
27	-5.314686e-05
28	-1.981959e-04

Table 8: Analytical Simpsons Sensitivity Model 2

7. Inference

1. From the numerical sensitivity of Trapezoidal vs Simpson's, it is inferred that the Simpson's approach with two points produces results with an error percentage of around 10 percent.
2. On comparing the computation time of Trapezoidal vs Simpson's approach, Simpson's approach is 10 times efficient than the trapezoidal approach.
3. On comparing the analytical linear end compliance with twice the increase in computational cost, the error can be decreased up to 50 percent.

8. References

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